

Exponential Functions & Growth Models

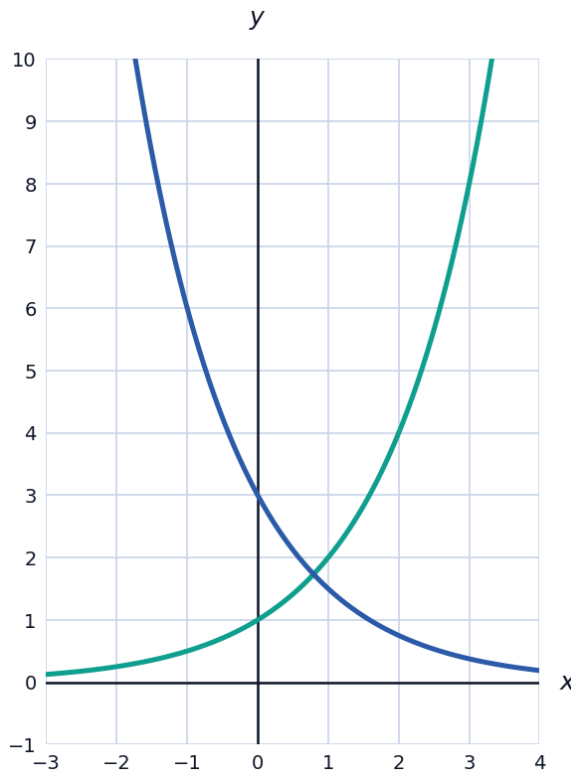


Exponential growth and decay basics

- 1 Classify each function as exponential growth or decay, and state the growth/decay factor: $f(x) = 5(1.2)^x$ and $g(x) = 5(0.8)^x$.
- 2 A population of bacteria is modeled by $P(t) = 1000(2)^{t/3}$, where t is in hours. Interpret the meaning of the exponent's structure, and find the population after 9 hours.

Visualizing exponential growth

- 3 The graph shows $f(x) = 2^x$ and $g(x) = 3(0.5)^x$, one growing and one decaying.



- a Identify which curve represents growth and which represents decay.
- b State the horizontal asymptote shared by both curves.

Finding an exponential model from data

- 4 An investment grows from 2000 dollars to 2420 dollars in 2 years under exponential growth $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

- 5 A radioactive sample decays from 80 g to 45 g over 6 years. Find the annual decay factor b in the model $A(t) = 80b^t$, to four decimal places.
-

Continuous growth and the number e

- 6 A quantity grows continuously according to $A(t) = 500e^{0.03t}$. Find the amount after 20 years, to the nearest whole number.
- 7 Explain the difference between the discrete growth model $A(t) = A_0(1 + r)^t$ and the continuous growth model $A(t) = A_0e^{kt}$, and state how r and k are related when both models describe the same annual growth rate r .
-

Comparing growth rates: exponential vs. polynomial

- 8 Explain why, for large enough x , any exponential growth function $f(x) = b^x$ (with $b > 1$) will eventually exceed any polynomial function $g(x) = x^n$, no matter how large the exponent n is. Illustrate with $f(x) = 1.01^x$ vs. $g(x) = x^{10}$.
-

Applications: half-life and doubling time

- 9 A substance has a half-life of 8 days. Write its decay model $A(t) = A_0\left(\frac{1}{2}\right)^{t/8}$, then find what fraction of the original amount remains after 20 days.
- 10 A city's population doubles every 15 years and currently stands at 40,000. Write an exponential model $P(t)$ and predict the population after 45 years.
-

Transformations of exponential functions

- 11 Describe how the graph of $g(x) = 2^{x-3} + 4$ is transformed from the parent function $f(x) = 2^x$, and state the new horizontal asymptote.
- 12 Find the horizontal asymptote and y -intercept of $h(x) = -3(2)^x + 6$.
-

Compound interest and financial models

- 13** A savings account earns 4% annual interest compounded quarterly, modeled by $A(t) = A_0 \left(1 + \frac{0.04}{4}\right)^{4t}$. If 5000 dollars is deposited, find the balance after 3 years, to the nearest dollar.
- 14** Explain, without computing exact values, why compounding more frequently (e.g. daily instead of quarterly) at the same nominal annual rate produces a slightly larger balance after the same number of years.